

Reusable equations for TRON–AIRBOT cooperative manipulation

Generalized momentum observer

$$M(\mathbf{q})\dot{\mathbf{v}} + C(\mathbf{q}, \mathbf{v})\mathbf{v} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau} + \boldsymbol{\tau}_{ext}, \quad (\text{OBS-01})$$

$$\boldsymbol{\tau}_{ext} = \mathbf{J}_e^T \mathbf{h}_e, \quad (\text{OBS-02})$$

$$\mathbf{p} = M(\mathbf{q})\mathbf{v}, \quad (\text{OBS-03})$$

$$\dot{\mathbf{p}} = \boldsymbol{\tau} + \boldsymbol{\tau}_{ext} - \boldsymbol{\beta}, \quad \boldsymbol{\beta} = \mathbf{g} - C^T \mathbf{v}, \quad (\text{OBS-04})$$

$$\mathbf{z}(t) = \int_0^t (\boldsymbol{\tau} - \boldsymbol{\beta} + \mathbf{r}) d\xi, \quad (\text{OBS-05})$$

$$\mathbf{r} = \mathbf{K}_O[(\mathbf{p} - \mathbf{p}_0) - \mathbf{z}], \quad (\text{OBS-06})$$

$$\dot{\mathbf{r}} = \mathbf{K}_O(\boldsymbol{\tau}_{ext} - \mathbf{r}). \quad (\text{OBS-07})$$

Bias compensation and wrench reconstruction

$$\alpha_b = 1 - e^{-\Delta t/T_b}, \quad (\text{BIAS-03})$$

$$\hat{\mathbf{b}}_k = \hat{\mathbf{b}}_{k-1} + \alpha_b(\bar{\mathbf{r}}_k - \hat{\mathbf{b}}_{k-1}), \quad (\text{BIAS-03b})$$

$$\hat{\boldsymbol{\tau}}_{ext,k} = \bar{\mathbf{r}}_k - \hat{\mathbf{b}}_k, \quad (\text{BIAS-04})$$

$$\hat{\mathbf{h}}_e = (\mathbf{J}_e \mathbf{J}_e^T + \lambda \mathbf{I})^{-1} \mathbf{J}_e \hat{\boldsymbol{\tau}}_{ext}. \quad (\text{WRE-03})$$

Grasp map and internal wrench

$$\mathbf{G}_i = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ [\mathbf{r}_i]_{\times} & \mathbf{I} \end{bmatrix}, \quad \mathbf{G} = [\mathbf{G}_1 \ \mathbf{G}_2], \quad (\text{GRASP-02})$$

$$\mathbf{w}_o = \mathbf{G}\mathbf{h}, \quad (\text{GRASP-03})$$

$$\mathbf{h}_{eff} = \mathbf{G}^+ \mathbf{w}_o, \quad \mathbf{h}_{int} = \mathbf{h} - \mathbf{h}_{eff}, \quad (\text{INT-01})$$

$$\mathbf{G}\mathbf{h}_{int} = \mathbf{0}. \quad (\text{INT-02})$$

Virtual-linkage internal-force coordinates

$$\begin{bmatrix} \mathbf{f}_r \\ \boldsymbol{\mu}_r \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{I} & \cdots & \mathbf{I} \\ [\mathbf{r}_1]_{\times} & \cdots & [\mathbf{r}_n]_{\times} \end{bmatrix}}_{\mathbf{G}_f} \mathbf{f}, \quad (\text{VL-02})$$

$$\mathbf{f} = \mathbf{B}_v \mathbf{t} + \mathbf{f}_e, \quad \mathbf{G}_f \mathbf{B}_v = \mathbf{0}, \quad (\text{VL-03})$$

$$\hat{\mathbf{t}} = \mathbf{B}_v^+ \mathbf{f}, \quad (\text{VL-04})$$

$$\mathbf{e}_{12} = \frac{\mathbf{p}_2 - \mathbf{p}_1}{\|\mathbf{p}_2 - \mathbf{p}_1\|}, \quad \mathbf{B}_{12} = \begin{bmatrix} \mathbf{e}_{12} \\ -\mathbf{e}_{12} \end{bmatrix}, \quad (\text{VL-05})$$

$$t_{12} = \frac{1}{2} \mathbf{e}_{12}^T (\mathbf{f}_1 - \mathbf{f}_2), \quad (\text{VL-06})$$

$$t_{12}^* = k_t(d_0 - d) - d_t \dot{d}. \quad (\text{VL-07})$$

For rigid 6D grasps, the complete internal-wrench dimension is

$$\dim \ker(\mathbf{G}) = 6(n - 1), \quad (\text{VL-09})$$

$$\mathbf{N}_G = \mathbf{I} - \mathbf{G}^+ \mathbf{G}, \quad (\text{VL-11a})$$

$$\mathbf{B}_{VL} = \text{orth}(\mathbf{N}_G \mathbf{B}_{raw}), \quad \mathbf{G} \mathbf{B}_{VL} = \mathbf{0}, \quad (\text{VL-11b})$$

$$\boldsymbol{\eta}_{VL} = \mathbf{B}_{VL}^+ \mathbf{h}_{int}. \quad (\text{VL-12})$$

Object impedance and load allocation

$$\mathbf{f}_o^* = \mathbf{K}_p(\mathbf{x}_o^* - \mathbf{x}_o) + \mathbf{D}_p(\dot{\mathbf{x}}_o^* - \dot{\mathbf{x}}_o) - \hat{m}_o \mathbf{g}, \quad (\text{LOAD-01})$$

$$\mathbf{h}^* = \mathbf{W}^{-1} \mathbf{G}^T (\mathbf{G} \mathbf{W}^{-1} \mathbf{G}^T)^+ \mathbf{w}_o^*. \quad (\text{LOAD-03})$$

Payload mass and center of mass

$$\boldsymbol{\theta}_o = [m, mc_x, mc_y, mc_z]^T, \quad (\text{PAY-01})$$

$$\mathbf{u} = \mathbf{a}_o - \mathbf{g}, \quad (\text{PAY-02})$$

$$\mathbf{w}_o = \underbrace{\begin{bmatrix} \mathbf{u} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 1} & -[\mathbf{u}]_{\times} \mathbf{R}_{WB} \end{bmatrix}}_{\mathbf{Y}_o} \boldsymbol{\theta}_o. \quad (\text{PAY-04})$$

CHIP-style hindsight compliance

$$\mathbf{g}_{hind} = \mathbf{g}_{ref} - \mathbf{C}_f \mathbf{f}_{ext}, \quad (\text{CHIP-01})$$

$$\boldsymbol{\tau} = \mathbf{g}(\mathbf{q}) + \mathbf{J}^T [\mathbf{K}_x(\mathbf{g} - \mathbf{x}) - \mathbf{D}_x \dot{\mathbf{x}}] + \mathbf{N}_J \boldsymbol{\tau}_{posture}, \quad (\text{CHIP-04})$$

$$\mathbf{K}_x \approx \mathbf{C}_f^{-1}, \quad (\text{CHIP-05})$$

$$\mathbf{N}_J = \mathbf{I} - \mathbf{J}^T (\mathbf{J} \mathbf{J}^T + \lambda_J \mathbf{I})^{-1} \mathbf{J}. \quad (\text{CHIP-06})$$

Object impedance: derivation and scope

Reading note: star2dust, *Impedance control in cooperative manipulation* (2020-08-17). Equations below are re-derived in the project convention. They are not a claim of implemented inertia shaping.

Translational target dynamics

For $\mathbf{e}_x = \mathbf{x} - \mathbf{x}_d$ and positive-definite, constant gains:

$$\mathbf{M}_d \ddot{\mathbf{e}}_x + \mathbf{D}_d \dot{\mathbf{e}}_x + \mathbf{K}_d \mathbf{e}_x = \mathbf{f}_{ext}, \quad (\text{IMP-01})$$

$$\mathbf{e}_{x,ss} = \mathbf{K}_d^{-1} \mathbf{f}_{ext}, \quad \mathbf{C}_f = \mathbf{K}_d^{-1}, \quad (\text{IMP-02})$$

$$\omega_n = \sqrt{k_d/m_d}, \quad \zeta = \frac{d_d}{2\sqrt{m_d k_d}}. \quad (\text{IMP-03})$$

At a fixed reference, the ideal energy balance is

$$\begin{aligned} V &= \frac{1}{2} \dot{\mathbf{e}}_x^T \mathbf{M}_d \dot{\mathbf{e}}_x + \frac{1}{2} \mathbf{e}_x^T \mathbf{K}_d \mathbf{e}_x, \\ \dot{V} &= \dot{\mathbf{e}}_x^T \mathbf{f}_{ext} - \dot{\mathbf{e}}_x^T \mathbf{D}_d \dot{\mathbf{e}}_x. \end{aligned} \quad (\text{IMP-04})$$

Object-level wrench command

World-frame twist is $\boldsymbol{\nu}_o = [\mathbf{v}_c^T, \boldsymbol{\omega}^T]^T$; the reference point is the center of mass.

$$\begin{aligned} \mathbf{M}_o \dot{\boldsymbol{\nu}}_o + \mathbf{b}_o &= \mathbf{w}_{ext} + \mathbf{G}\mathbf{h}, \\ \mathbf{M}_o &= \text{diag}(m_o \mathbf{I}_3, \mathbf{I}_W), \\ \mathbf{b}_o &= \begin{bmatrix} -m_o \mathbf{g} \\ \boldsymbol{\omega} \times (\mathbf{I}_W \boldsymbol{\omega}) \end{bmatrix}, \quad \mathbf{I}_W = \mathbf{R}_{WB} \mathbf{I}_B \mathbf{R}_{WB}^T. \end{aligned} \quad (\text{IMP-05})$$

The following linear pose-error form requires translation or consistent local small-angle coordinates, not raw Euler-angle derivatives:

$$\mathbf{a}_{imp} = \mathbf{a}_{ref} + \mathbf{M}_d^{-1} (\mathbf{w}_{ext} - \mathbf{D}_d \mathbf{e}_\nu - \mathbf{K}_d \mathbf{e}_o). \quad (\text{IMP-06})$$

$$\begin{aligned} \mathbf{w}_{cmd} &= \mathbf{b}_o + \mathbf{M}_o \mathbf{a}_{ref} - \mathbf{M}_o \mathbf{M}_d^{-1} (\mathbf{D}_d \mathbf{e}_\nu + \mathbf{K}_d \mathbf{e}_o) \\ &\quad + (\mathbf{M}_o \mathbf{M}_d^{-1} - \mathbf{I}_6) \mathbf{w}_{ext}. \end{aligned} \quad (\text{IMP-07})$$

$$\mathbf{h}_{cmd} = \mathbf{G}^+ \mathbf{w}_{cmd} + \mathbf{B}_{VL} \boldsymbol{\eta}^*, \quad \mathbf{G} \mathbf{B}_{VL} = \mathbf{0}. \quad (\text{IMP-08})$$

Exact wrench realization requires a feasible request and adequate grasp-map rank. Here $\mathbf{w}_{ext}$ excludes robot contact wrenches and already modeled gravity. See Schneider and Cannon (1992) for the object impedance framework; the Markdown notes contain source links and caveats.

Local endpoint impedance: fixed-base derivation

$\mathbf{f}_i$ denotes force from robot to object, so the reaction on the robot is $-\mathbf{f}_i$. All endpoint equations here are translational.

$$\mathbf{M}_{d,i}\ddot{\mathbf{x}}_i + \mathbf{D}_{d,i}(\dot{\mathbf{x}}_i - \dot{\mathbf{x}}_{d,i}) + \mathbf{K}_{d,i}(\mathbf{x}_i - \mathbf{x}_{d,i}) = -\mathbf{f}_i. \quad (\text{DIMP-01})$$

For a fixed reference and zero initial conditions,

$$\mathcal{F}_i(s) = (\mathbf{M}_{d,i}s^2 + \mathbf{D}_{d,i}s + \mathbf{K}_{d,i})\mathcal{E}_i(s), \quad \boldsymbol{\epsilon}_i = \mathbf{x}_{d,i} - \mathbf{x}_i. \quad (\text{DIMP-02})$$

$$\mathbf{x}_{d,i} = \mathbf{x}_i + \mathbf{K}_{d,i}^{-1}\mathbf{f}_i \quad (\text{static equilibrium}), \quad (\text{DIMP-03})$$

$$\mathbf{x}_{d,i}^{cmd} = \mathbf{x}_i^{nom} + \mathbf{K}_{d,i}^{-1}\mathbf{f}_i^*. \quad (\text{DIMP-04})$$

Operational-space dynamics

Assume $\mathbf{J}$ has full row rank, and let $\mathbf{c}_q = \mathbf{C}_q\dot{\mathbf{q}}$.

$$\boldsymbol{\Lambda} = (\mathbf{J}\mathbf{M}_q^{-1}\mathbf{J}^T)^{-1}, \quad \bar{\mathbf{J}} = \mathbf{M}_q^{-1}\mathbf{J}^T\boldsymbol{\Lambda}. \quad (\text{DIMP-05})$$

$$\begin{aligned} \boldsymbol{\Lambda}\ddot{\mathbf{x}} + \boldsymbol{\mu}_x + \mathbf{p}_x &= \mathbf{F}_{cmd} - \mathbf{f}_i, \\ \boldsymbol{\mu}_x &= \boldsymbol{\Lambda}(\mathbf{J}\mathbf{M}_q^{-1}\mathbf{c}_q - \dot{\mathbf{J}}\dot{\mathbf{q}}), \\ \mathbf{p}_x &= \boldsymbol{\Lambda}\mathbf{J}\mathbf{M}_q^{-1}\mathbf{g}_q, \quad \mathbf{F}_{cmd} = \bar{\mathbf{J}}^T\boldsymbol{\tau}. \end{aligned} \quad (\text{DIMP-06})$$

The negative Jacobian-derivative term follows directly from $\ddot{\mathbf{x}} = \mathbf{J}\ddot{\mathbf{q}} + \dot{\mathbf{J}}\dot{\mathbf{q}}$. Define $\mathbf{s}_i = \mathbf{D}_{d,i}(\dot{\mathbf{x}}_i - \dot{\mathbf{x}}_{d,i}) + \mathbf{K}_{d,i}(\mathbf{x}_i - \mathbf{x}_{d,i})$. Then

$$\mathbf{F}_{cmd,i} = \boldsymbol{\mu}_{x,i} + \mathbf{p}_{x,i} - \boldsymbol{\Lambda}_i\mathbf{M}_{d,i}^{-1}\mathbf{s}_i + (\mathbf{I} - \boldsymbol{\Lambda}_i\mathbf{M}_{d,i}^{-1})\mathbf{f}_i. \quad (\text{DIMP-07})$$

$$\boldsymbol{\tau}_i = \mathbf{J}_i^T\mathbf{F}_{cmd,i} + (\mathbf{I} - \mathbf{J}_i^T\bar{\mathbf{J}}_i^T)\boldsymbol{\tau}_{0,i}. \quad (\text{DIMP-08})$$

For an error-acceleration impedance, add reference-acceleration feedforward. This derivation is for ideal fixed-base torque control without additional constraint contacts; it is not a floating-base walking controller. Operational-space background: Khatib (1987).